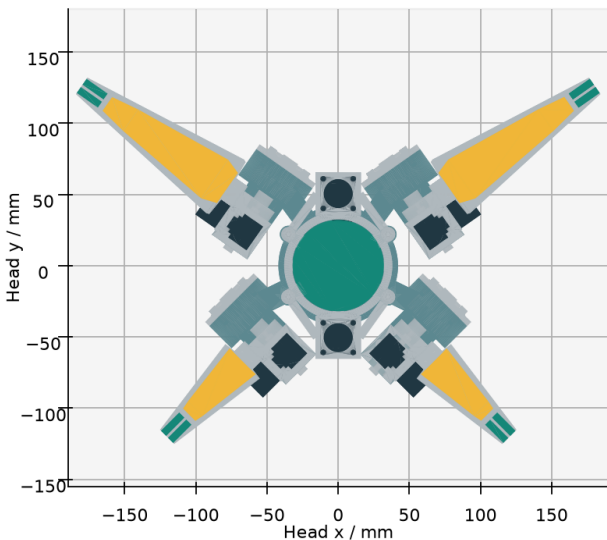
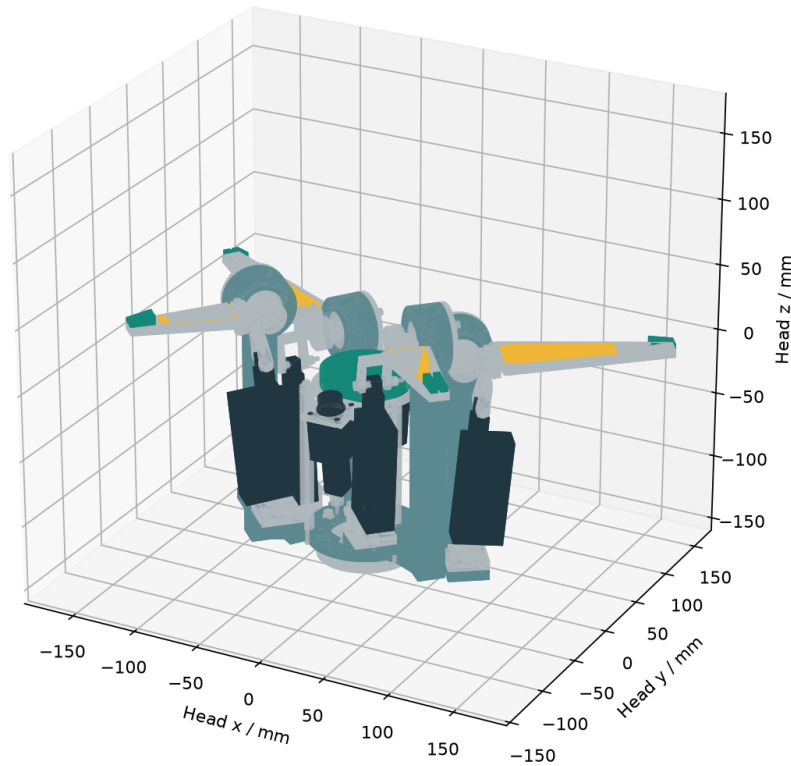


Carrier, four independent axes and removable optics

P16-CARRIER-01 | Load path and optical packaging study
Nominal metal geometry + original supplier envelopes; not manufacturing release
Original solid assembly / open pose Front / upper and lower pairs remain independent



Coordinates / assembly

Face origin: [0, 0, 0]; +Z forward.

J7 output contact: Z = -140.

Existing 10 mm adapter front: Z = -130.

Carrier rear-most metal: Z = -154.

Upper / lower finger roots: Z = 0 / +50.

Root radial distance: R70.

Upper q: 0..109 deg; lower: 0..122 deg.

Wrist blocked: q7=-90 hits J6.

Holding/payload are NOT qualified.

P16/camera shapes shown are envelopes.

Vendor CAD is not redistributed.

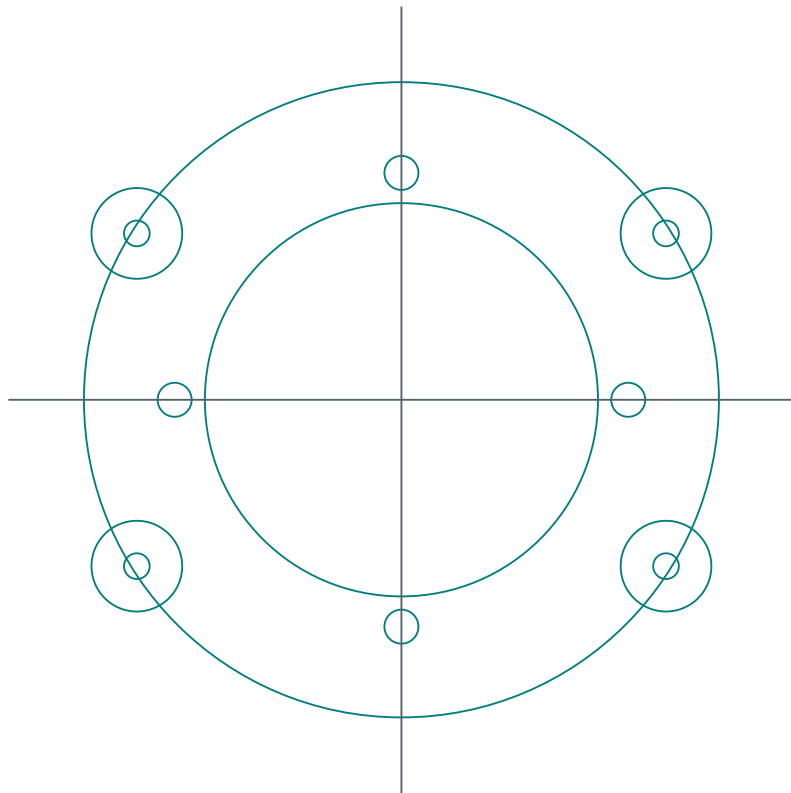
Assembly sequence

1. Attach J7 adapter; thread depth pending.
2. Attach carrier with four M4 candidates.
3. Install bearing pairs and root spindles.
4. Fit P16 pin/spacer stacks and base bolts.
5. Fit removable optical module last.

Authoritative nominal geometry: main-carrier.step, integral-steel-root-spindle.step, bearing-outer-cap.step, base-bracket.step and removable-optical-frame.step. Exact source hashes and QA are in study.json / review.json.
Alloy, fits, fillets, fatigue, bearing adjustment, fastener preload and cable strain relief remain engineering gates.
Full distal LED stack and mechanically retained soft pads are not integrated in this root/carrier branch.

Main carrier / J7 and structural section dimensions

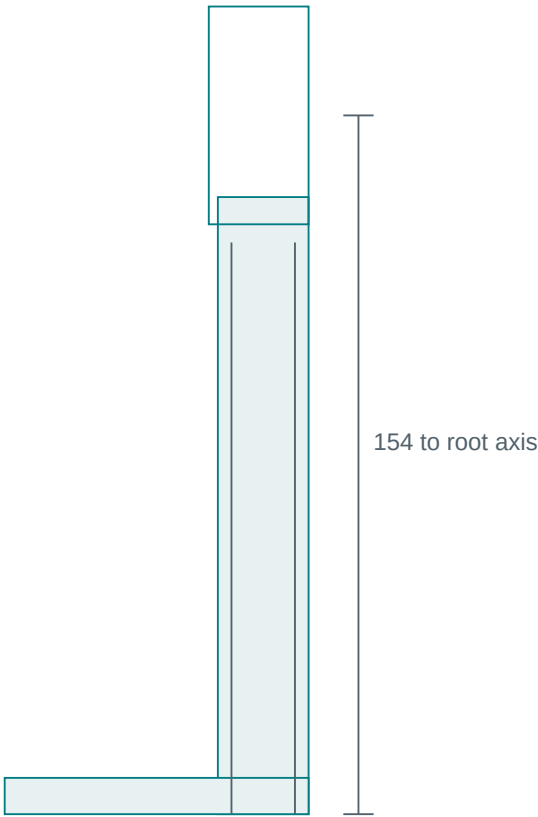
Carrier rear ring / XY / nominal 1:1



Ring: OD84 / ID52 / thickness8; Z -130..-122.
Four D4.5 clearance holes: (30,0),(0,30),(-30,0),(0,-30).
Four optical-column bosses: (+/-35,+/-22), OD12 / holeD3.4.
Radial webs: width16 x thickness8. Full silhouette: STEP.
Module stems: 20 x 20 closed section, blind D14 bore.
Minimum nominal straight wall3; end ligament4 below housing.

J7 adapter uses the measured 8-hole PCD44 pattern, NOT uniform 45 degree spacing.
Angles: 52.64583, 82.64583, 142.64583, 172.64583, 232.64583, 262.64583, 322.64583, 352.64583 deg.
J7 screw length remains TBD: manufacturer recess9 / effective thread5 does not identify full thread runout.

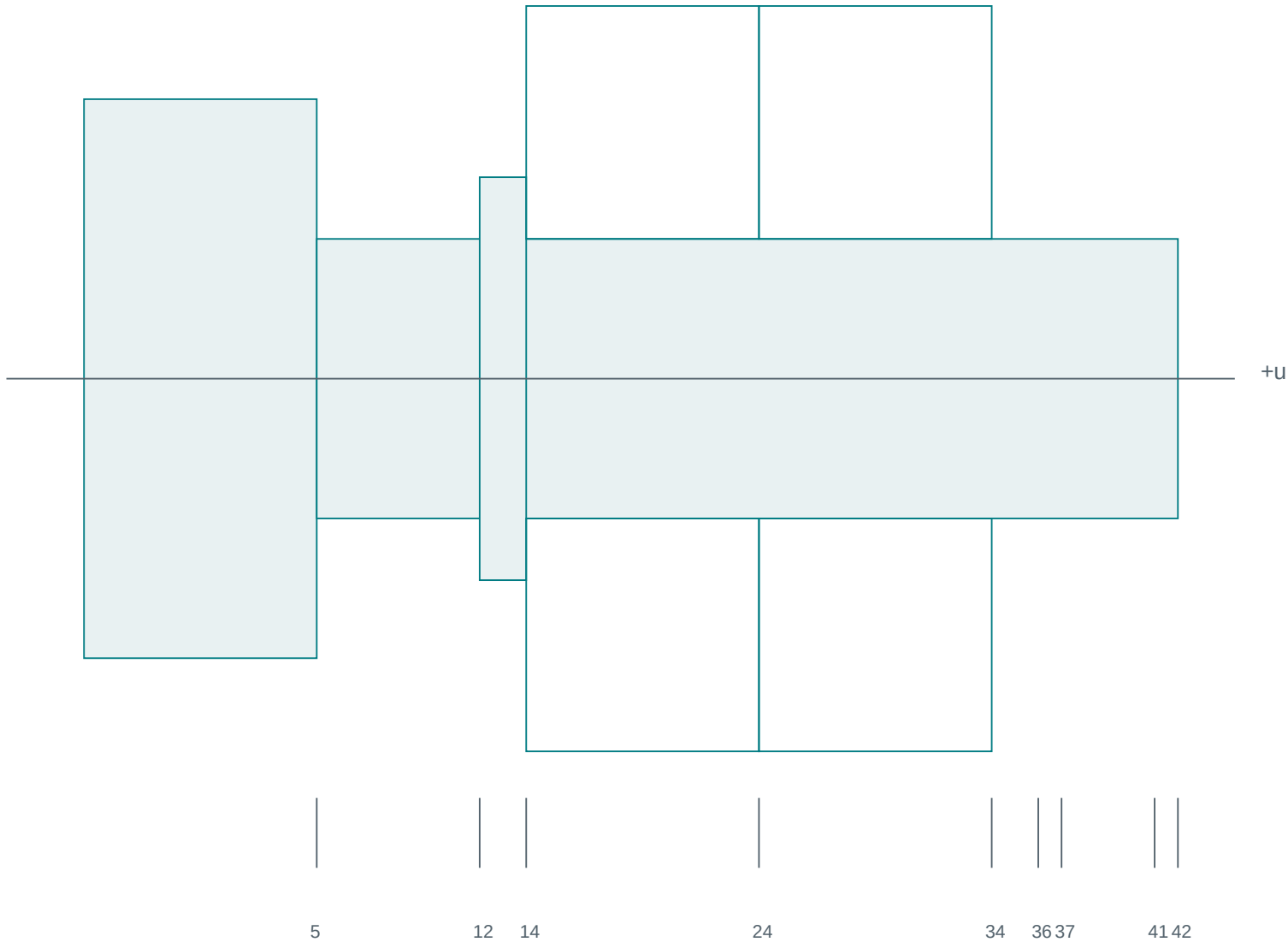
Upper module / u-Z schematic / 0.6:1



Upper bore length126; lower152 (root Z offset+50).
Base arm Z(local) -154..-146; bracket -146..-139.
Base M4 holes: local (r,u) = (+/-10,-18).
Counterbores D7.5 x 4.2; through D4.4.
Nominal M4x10: 7.2 engagement; not torque release.
A single machined solid is geometrically connected.
Deep boring, tool access and distortion need DFM.

Integral root spindle and double bearing cartridge

Bearing section / selected dimensions enlarged 3.6:1



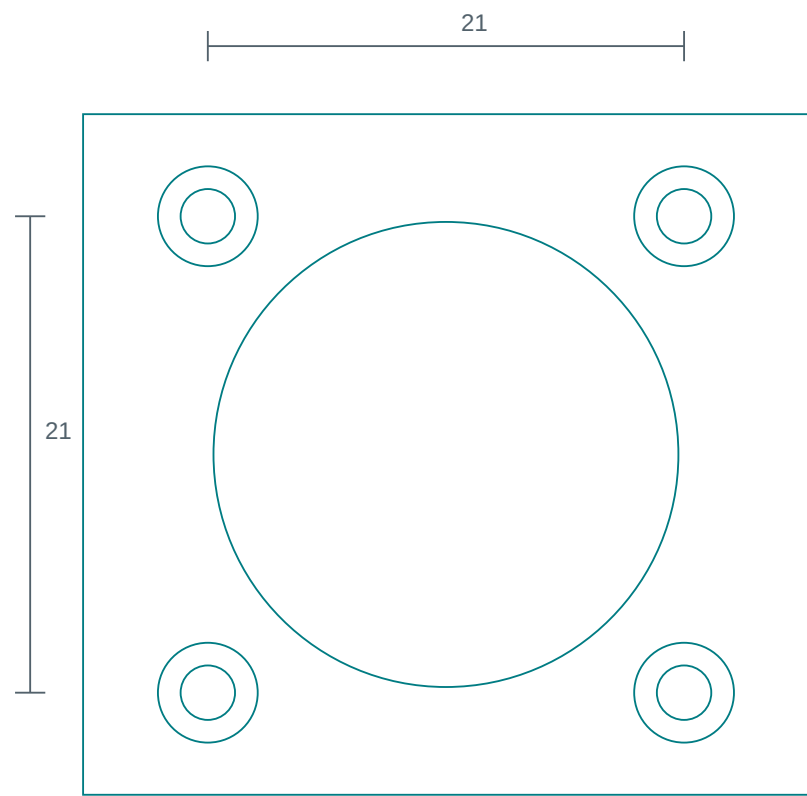
Two SKF 7201 BECBP: 12 x 32 x 10.
Bearing B: u14..24; A: u24..34.
Rear inner shoulder: OD17.3,u12..14.
Rear outer shoulder: boreD28,u12..14.
Carrier body OD48,u12..34.
Cap OD48,u34..38.
Cap boreD27.5 at u34..36;
front reliefD30 at u36..38.
4 x M3 on PCD40, angles45+90k.
Cap throughD3.4, body threadM3.
M3x10 / washer0.5; 5.5 engagement.

Lock end: spacer u34..36, MB1 u36..37,
KM1 M12x1 u37..41; end at u42.
MB1 slot nominal3.2 wide x1.6 deep,
u35.8..42.1; no internal axial M4.
Thread/MB1 bent tabs are simplified.
Fit, thread runout and washer to verify.

One steel part carries crank -> narrow blade fork and integral solid bearing spindle; no keyed D10 tenon.
P16 tip pin center at q0: (r,u,z) = (9.739771,-18,-24.106780); crank radius26; phase68 degrees.
P16 pin gaps: tip14.2 / base22; cheeks4; real NBK D4 shoulders with M3 threads outside both cheeks.
Do not infer preload: BECBP has catalog CB positive clearance. Outer-ring adjustment and actual pressure centers are unresolved.
Own support-shell clearance: upper0.226 / lower0.380. No tolerance allowance; near-root transition needs relief.

Front camera interface and removable optical module

Original camera plate / 3:1 / integrated in front frame



Camera native mounting plane z0.
32 x 30 plate, thickness4.
Lens clearanceD20.5.
4 x throughD2.4 / counterboreD4.4 x2.
M2x6 head below front plane;
nominal body engagement4 of available6.
Countersink is NOT used.

Camera poses: (0,+/-50,-14) mount centers.
Pitch outward8 deg about head X.
Native front lens extends to z+13.911658.
Physical back extends to z-33.550037.
Total physical CAD depth47.461695.
Summary34.2 excludes rear hardware.

D64 x12 central display reservation:
head Z -11..+1, includes GH allowance.
No routed round PCBA is claimed.
Keepout is excluded from hardware mass.

Removable optical frame Z around -15..-11, with camera plates tilted8 degrees; exact relief silhouette is STEP.
Four columns OD10 / ID3.4 / L107, centers(+/-35,+/-22), Z-122..-15. Four cut M3 tie rods, L128.
Tie rods are nominal thread envelopes with two nuts/washers each; final locking and compression stiffness TBD.
Actual camera STEP contains Image plane and Simple FOV reference solids; these are not mechanical material.
Female FAKRA reserved at full unattached26.2 mm length behind male; no precise mated datum is asserted.
Lens FOV occlusion, cable bend radius, strain relief and connector withdrawal need follow-on optical/harness validation.